The generating loop of the circle is not the reflexivity path

The Univalent Foundations Program

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Editorial note. The counted statement is the author lem environment at source lines 444-446 and its proof environment opens on line 447, the next line; the pairing is by adjacency and the retained proof range 447-452 is the whole environment.

Editorial note. The retained counter values (chapter 6, section 4, theorem offset 0) make the counted lemma print as Theorem 6.4.1, matching the original book numbering read from the label-number dump of the pinned snapshot (newlabel thm:loop-nontrivial = 6.4.1).

Editorial note. One label the proof cites lies outside the card, thm:type-is-not-a-set (book Example 3.1.9, defined at line 133 of the logic chapter pinned file), and is declared in external_references as a non-load-bearing navigation pointer: the retained proof states the contradiction it needs in its own words.

Editorial note. The retained preamble is the book own macro library and layout adapter as stored for this chapter; no package outside it is requested.

Editorial scope. One original result of the pinned HoTT Book source tree, counted once: the generating loop of the circle is not trivial (author label thm:loop-nontrivial, book Theorem 6.4.1).

Allowed standard prerequisites (editorial).

- The circle as a higher inductive type. The circle has a point constructor and a loop constructor, so a map out of it into any type with a point x and a loop p at x is given by those two data.. It is what turns a hypothetical triviality of the loop into an equation in an arbitrary type.
- The universe is not a set. Not every type is a set: the universe contains a type whose loop space has two distinct elements.. It is the contradiction the proof arrives at, and it is retained as a navigation pointer.

Source reference. Original remark 3.1.9: sources/hott-logic/pinned-original-source.tex, source lines 133–133. The book Example 3.1.9, that the universe is not a set, invoked at source line 451 as the contradiction that closes the argument. Navigation pointer only: the retained proof states the step it needs and is complete; the pointer span is the label line of the pinned file that carries that example.

Lemma 6.4.1. $\mathsf{loop} \neq \mathsf{refl}_{\mathsf{base}}$.

Proof. Suppose that $\mathsf{loop} = \mathsf{refl}_{\mathsf{base}}$. Then since for any type A with $x : A$ and $p : x = x$, there is a function $f : \mathbb{S}^1 \to A$ defined by $f(\mathsf{base}) :\equiv x$ and $f(\mathsf{loop}) \coloneqq p$, we have

$$p = f(\mathsf{loop}) = f(\mathsf{refl}_{\mathsf{base}}) = \mathsf{refl}_x.$$

But this implies that every type is a set, which as we have seen is not the case (see original remark 3.1.9). □